BTS-V(15)-11.17-1524 Reg.No. [| | | | | | | |]

B

B.Tech. Degree V Semester Examination November 2017

ME 15-1502 MECHANICS OF MACHINERY
(2015 Scheme)

Time : 3 Hours

Maximum Marks : 60

PART A
(Answer *ALL* questions)

$(10 \times 2 = 20)$

I. (a) Explain any one intermittent motion mechanism with a neat figure.

(b) Derive an expression for the ratio of times taken in forward and return strokes in a *Crank and Slotted Lever* Quick Return Motion Mechanism.

(c) Explain any two inversions of a four bar mechanism with a neat sketch.

(d) Define the terms (i) module (ii) pressure angle (iii) circular pitch (iv) addendum.

(e) Derive an expression for the minimum number of teeth required on the pinion in order to avoid interference in involute gear teeth.

(f) Derive an expression for displacement and velocity for a tangent cam operating on a radial-translating roller follower, when the contact is on the straight flank.

(g) Derive an expression for the frictional torque in flat pivot friction considering uniform pressure condition.

(h) Explain the method of obtaining the co-ordinates of a coupler point in a four bar mechanism.

(i) Explain any two types of dynamometers with neat sketches.

(j) Calculate the maximum and minimum transmission angles for a slider crank mechanism.

PART B

$(4 \times 10 = 40)$

II. (a) The angle between the axes of two shafts connected by Hooke's joint is 18°. Determine the angle turned through by the driving shaft when the velocity ratio is (i) maximum and (ii) unity. (4)

(b) The crank and connecting rod of a horizontal steam engine are 0.5 m and 2 m long respectively. The crank makes 180 rpm in the clockwise direction. When it has turned 45° from the inner dead centre position, determine (i) velocity of piston (ii) angular velocity of connecting rod (iii) velocity of a point on the connecting rod 1.5 m from the gudgeon pin. (6)

OR

(P.T.O.)

III. (a) In a Davis steering gear, the distance between the pivots of the front axle is 1.2 m and the wheel base is 2.7 m. Determine the inclination of the track arm to the longitudinal axis of the car, when it is moving along a straight path. (4)

(b) A link AB of a four bar linkage $ABCD$ revolves uniformly at 120 rpm in a clockwise direction. Determine the angular accelerations of links BC and CD and acceleration of point E in link BC. Given :
$AB = 7.5\ cm,\ BC = 17.5\ cm,\ EC = 5cm,\ CD = 15\ cm,\ DA = 10\ cm$
and Angle BAD = 90°. (6)

IV. A pair of 20° full depth involute spur gears having 30 and 50 teeth respectively of module 4 mm are in mesh, the smaller gear rotates at 1000 rpm. Determine (i) sliding velocities at engagement and at disengagement of a pair of teeth (ii) contact ratio. (10)

OR

V. An epicyclic gear train consists of a sun wheel S, a stationary internal gear E and three identical planet wheels P carried on a star shaped planet carrier C. The size of different toothed wheels is such that the planet C rotates at 1/5 of the speed of the sun wheel S. The minimum number of teeth on any wheel is 16. The driving torque on the sun wheel is 100 Nm. Determine (i) the number of teeth on different wheels of the train (ii) torque required to keep the internal gear stationary. (10)

VI. From the following data draw the profile of a cam in which the follower moves with SHM during ascent while it moves with uniform acceleration and deceleration during descent. (10)

Lift of follower	=	$4\ cm$
Least radius of cam	=	$5\ cm$
Angle of ascent	=	48°
Angle of dwell between ascent and descent	=	42°
Angle of descent	=	60°
Diameter of roller	=	$3\ cm$
Distance between line of action of follower and the axis of the cam	=	$2\ cm$

If the cam rotates at 360 rpm anticlockwise, find the maximum velocity and acceleration of the follower during descent.

OR

VII. (a) Derive an expression for the efficiency of a screw jack. (3)

(b) The pitch of 50 mm mean diameter threaded screw of a screw jack is 12.5 mm. The coefficient of friction between the screw and the nut is 0.13. Determine the torque required on the screw to raise a load of 25 kN, assuming the load to rotate with the screw. Also determine the efficiency of the machine. (7)

VIII. Determine the lengths of all the four links in a four bar chain for the length of the smallest being 10 cm to generate $y = log_{10}\ x$ in the interval $1 \leq x \leq 10$ for three accuracy points. The range of angles of input link and output link are $45° \leq \theta \leq 105°, 135° \leq \phi \leq 225°$ respectively. (10)

OR

IX. A single plate clutch, effective on both sides, is required to transmit 25 kW at 3000 rpm. Determine the outer and inner radii of the frictional surface if the coefficient of friction is 0.255, the ratio of radii is 1.25 and the maximum pressure is not to exceed 0.1 N/mm^2. Also determine the axial thrust to be provided by springs. Assume uniform wear theory condition. (10)

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